

Prognostic Factors for Extra-Abdominal and Abdominal Wall Desmoids: A 20-Year Experience at a Single Institution

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Background and Objective: Previous reports even large studies discussing the prognosis of desmoids have included tumors from intra- and extra-abdominal sites as well as incomplete resection. The purpose of this study was to explore prognostic factors associated with the recurrence free survival (RFS) rate in surgically treated extra-abdominal and abdominal wall desmoids.

Patients and Methods: A total of 198 consecutive desmoid patients were treated with surgery over a 20-year period at a single institution. Of these, 151 patients with extra-abdominal and abdominal wall tumors were retrospectively reviewed. One hundred thirteen patients were referred for the primary tumor and the other 38 for recurrent disease initially treated elsewhere. All patients underwent a macroscopically complete resection.

Results: The median follow-up interval was 102 months. Thirty-one patients (20.5%) had a local recurrence (LR). No patients died of the disease. The 5- and 10-year RFS was 79.7% and 78.5%, respectively. Admission status, gender, tumor size, margin status, location, and number, were predictors of LR in univariate analysis. Tumor size and margin status were independent prognostic factors in multivariate analysis. Positive margins were predictive of recurrence of primary disease, and also showed a trend for recurrent disease, which was not statistically significant. The selective use of adjuvant radiation did not show significant benefit over local control.

Conclusions: Regardless of primary or recurrent disease, microscopically negative margins should always be the goal for extra-abdominal desmoids surgery, if no cosmetic defects or function demolition is encountered. Extra-abdominal desmoids deserve more attention and should be treated more aggressively, especially when leaving positive margins.

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KEY WORDS: desmoids; prognostic factors; surgery; recurrence

INTRODUCTION

Desmoid tumors (DTs), also called aggressive fibromatosis, are rare neoplasms that arise from fascial or deep musculo-aponeurotic structures. The worldwide incidence rate of these tumors is low, at about 2–4 per million per year [1], and they account for about 0.03% of all neoplasms and <3% of all soft tissue tumors [2]. Despite their benign appearance and lack of any propensity to metastasize, they display a local aggressiveness to surrounding structures that enhances the difficulty of resection and that may cause serious clinical problems. Desmoids can arise sporadically or in association with familial adenomatous polyposis (FAP), which may hold a higher incidence rate of 3.5–32% [3]. Although the exact etiology of desmoids is unknown, genetic abnormalities, trauma, and steroid sex hormones may contribute to their oncogenesis [4].

Desmoids can occur anywhere in the body. According to their distribution, they can be classified as extra-abdominal, abdominal wall, and intra-abdominal types. All of these have their own features, including a different growth pattern, age predilection, and recurrence rates [5,6]. Due to their unpredictable and enigmatic clinical behavior, the effects of treatment are further confounded by some tumors that apparently regress or remain stable, even without treatment [7].

Optimal management has not yet been established or even clearly defined. Multidisciplinary management is now receiving increased attention and emphasis, to include surgery, radiotherapy and systemic therapy. Surgery remains the mainstay of treatment. However, despite wide surgical resection, there is still a relative high local recurrence (LR) rate that varies among the different anatomical subgroups, at 24% in the abdominal wall group, 43% in the extra-abdominal, and 77% in the intra-abdominal group [5].

Because of its rarity and variable natural history, no randomized trials have yet been carried out on the therapy of this tumor. Although

many retrospective and comparative analysis studies have been done, controversy still exists, especially regarding the impact of microscopic margins status, as well as the effects of radiation therapy. However, authors throughout the literature always presented data mixing anatomical subgroups and combined neoadjuvant or adjuvant therapy, as well as primary or recurrent tumors. To assess the value of prognosis factors in primary and recurrent tumors, and particularly the impact of microscopic margin status, we conducted a retrospective analysis of patients treated at our institution over a 20-year period.

METHODS AND PATIENTS

Methods

Between 1987 and 2007, 151 patients with pathologically diagnosed DTs, who underwent macroscopically completed resection, were recorded and retrospectively analyzed. All had been operated on with eradicating intent and had been followed up until July 2008. Patients with intra-abdominal diseases and Gardner syndrome, as well as with macroscopically uncompleted resection, were excluded from the study. The dates of initial treatment and LR were recorded.

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Patients

Patient demographics and clinical parameters were retrospectively obtained from patient charts. Patient age was defined as the age at presentation to FUSCC, and it was categorized as ≤ 30 years or >30 years for analysis. Tumor size was determined by pathology reports and was split into ≤ 5 cm or >5 cm for analysis. Tumor sites were categorized as an abdominal wall group and an extra-abdominal group including extremities, girdles, head and neck, or chest wall and back. Multifocality was defined pathologically for synchronous lesions; however, metachronous lesions were excluded as these were not LR's.

Microscopic margin status was retrieved from final pathology reports and retrospectively reviewed by a single pathologist. Radiation therapy was delivered by electron beam at doses from 45 to 55 Gy to selected patients based on the decision of the operating surgeon in the adjuvant setting. The indication for radiation therapy included when marginal resection was undergone or when a higher risk of recurrence was predicted. Systematic therapy was most commonly tamoxifen, prescribed by operating surgeon to selected patients at 30 mg/day, while chemotherapy that included doxorubicin-based regimens was delivered for one patient in a neoadjuvant setting.

Local failure and RFS were calculated from the time of operation. Recurrence was defined as the local appearance of a new lesion after a macroscopically complete resection. RFS was calculated using the Kaplan–Meier Method and Rothman's 95% confidence intervals (95% CI). Univariate and multivariate analysis were performed using the log-rank test and the Cox models for prognosis factors of outcome. A P -value <0.05 was considered significant. Analyses were performed using SPSS statistical software version 15.0.

RESULTS

Patient Characteristics

Between January 1987 and December 2007, 198 patients diagnosed with DT were treated at the Fudan University Shanghai Cancer Centre (FUSCC), China. Of these, 12 were classified as intra-abdominal desmoids, including 3 patients with Gardner syndromes, and were all excluded from the study. Another 35 extra-abdominal or abdominal wall patients who received biopsies or macroscopically incomplete resections or who developed multiple recurrent diseases were also excluded. The outcomes of the remaining 151 patients formed the base of this article (Table I).

Median age of patients at the time of initial diagnosis was 34 years (range, 7–86 years). One hundred nineteen patients were female, and 32 were male. The male to female ratio was approximately 1:3.7. Tumors were located at the following sites: anterior abdominal wall in 75 patients, and extra-abdominal in 76 patients including extremities in 19 patients (lower/upper limbs in 15/4), girdles in 14 patients (scapular/pelvic girdle in 5/9), anterior chest wall/back in 9/8 patients, and head and neck in 26 patients. Tumor size ranged from 0.8 to 25 cm (≤ 5 cm in 73 patients, ≤ 10 cm in 55 patients, ≤ 15 cm in 16 patients, and ≥ 15 cm in 7 patients). The median size of tumor for the entire group was 55 mm. Fourteen patients were affected by multicentric tumors (nine patients with two tumors, five patients with three tumors). In all, one hundred and thirteen patients were referred with primary disease, whereas the other 38 patients experienced a recurrent tumor initially treated elsewhere.

Patient Outcomes

The median follow-up interval was 102 months (range, 7–256 months). Twelve patients were lost at follow-up postoperatively, with the median time of 32.5 months (range, 7–99 months). No deaths

TABLE I. Factors Potentially Affecting Local RFS (All 151 Patients)

Factors	No	10-year RFS		
		%	95% CI	P
Admitted status				
Primary	113	85	78–92	0.001
Recurrent	38	56	21–73	
Age (years)				
≤ 30	62	75	63–87	0.587
>30	89	81	72–89	
Sex				
Female	119	82	75–89	0.025
Male	32	63	43–83	
Tumor size (cm)				
≤ 5	73	93	87–99	0.000
>5	78	63	51–75	
Site				
Extra-abdominal	77	68	57–79	0.001
Extremity/girdles	33	69		
Chest wall/back	17	76		
Head/neck	26	59		
Abdominal wall	75	89	81–96	
Number				
Single	137	83	76–89	0.000
Multicentric	14	18	0–48	
Margin				
Negative (R0)	106	88	81–94	0.000
Positive (R1)	45	54	38–70	
Adjuvant radiation				
No	126	81	74–88	0.062
Yes	25	63	43–83	
Tamoxifen therapy				
No	126	81	73–88	0.174
Yes	25	68	48–88	

were observed. One patient experienced a radiation-related secondary peripheral nerve sheath tumor (MPNST). In total, 31 patients developed 42 LR's: 24 patients had 1 recurrence, 4 patients had 2, 2 patients experienced 3 recurrences, and 1 suffered 4 recurrences. The first LR rate was 31/151 (21%). Time to first recurrence varied from 4 to 167 months; the median time for initial recurrence was 16 months. The recurrence-free survival rate (RFS) was 79% (95% confidence interval [CI], 72–86%) at 5 years and 78% (95% CI, 71–85%) at 10 years. No patient had metastatic disease.

Overall, 31 recurrences were treated either by surgery alone ($n=6$), surgery combined with an additional modality ($n=8$), radiation therapy ($n=12$), or by observation ($n=5$). Of these, 12 were free of disease, and 12 obtained stable disease during the follow-up interval. Seven suffered a second recurrence and were treated with surgery combined with additional therapies ($n=3$), radiation alone ($n=2$), or systemic therapy alone ($n=2$). Four were free of disease after 48, 35, 17, and 51 months of follow-up. Three patients have had three or more recurrences.

Prognostic Factors on RFS

Univariate analysis identified admission status, gender, tumor size, margin status, location, and number as significantly correlated with LR (Table I), while size and margin status were independent prognostic factors on multivariate analysis (Table II).

Patients with primary lesions at first surgery in our institution had a better outcome, with a 5-year RFS rate of 87% (95% CI, 81–92%) and 85% (95% CI, 78–91%) at 10 years. However, for recurrent tumors, it was 56% (95% CI, 40–72%) and unchanged at 10 years. The difference in outcome depending on previous surgical history was statistically significant ($P=0.001$, Fig. 1). We therefore analyzed

TABLE II. Multivariate Analysis of the Effect of Prognostic Factors on RFS (All Patients)

Factors	Hazard ration	95% CI		P
Admitted status (primary vs. recurrent)	1.60	0.66	3.88	0.295
Gender (male vs. female)	1.48	0.58	3.78	0.411
Tumor size (>5 vs. ≤5)	2.77	1.13	6.79	0.026
Location (extra-abdominal vs. abdominal)	0.46	0.19	1.13	0.089
Number (multicentric vs. single)	1.57	0.52	4.68	0.422
Margin (negative vs. positive)	3.11	1.40	6.93	0.005

gender, tumor sites, size, number and marginal status separately by subgroup.

For primary tumors, patients with tumors <5 cm, located in abdominal region, and attaining negative margins had a statistically better outcome than did those with tumors above 5 cm, located in extra-abdominal region, and experiencing positive margins. However, for recurrent disease, that tumor number was the only positive prognostic factor for LR ($P = 0.015$) (Table III).

Tumor Sites

Patients with abdominal wall tumors obtained a significantly better outcome, with a 5-year RFS rate of 91% (95% CI, 84–98%) and 89% (95% CI, 81–97%) at 10 years, whereas extra-abdominal tumors had a 5-year RFS rate of 67% (95% CI, 56–78%; $P = 0.001$) and were unchanged at 10 years.

For extra-abdominal tumors, different subgroups also held different outcomes. Tumors in the chest wall and back had a relatively better outcome, with a 5-year RFS of 76% (95% CI, 55–97%) that was unchanged at 10 years, while tumors in the extremities or girdles had a 5-year RFS of 68% (95% CI, 51–85%) that was unchanged at 10 years. Head and neck tumors had the worst outcomes, with a 5-year RFS of 59% (95% CI, 40–78%) that was unchanged at 10 years (Fig. 2).

Resection Margins

Marginal status was an independent predictive factor of LR. For primary disease, patients with positive margins had a 5-year RFS of

64% (95% CI, 44–83%) that was unchanged at 10 years, whereas those with negative margins had a 5-year RFS of 92% (95% CI, 85–98%; $P = 0.000$).

For recurrent disease, there was a trend towards predicting local progression for positive margins, although this was not statistically significant. Patients with positive margins had a 5-year RFS of 35% (95% CI, 13–57%) that was unchanged at 10 years, whereas those with negative margins had a 5-year RFS of 71% (95% CI, 52–89%) that was unchanged at 10 years ($P = 0.09$, Fig. 3).

Considering that combined adjuvant therapy might bias the results, we further investigated the impact of margin status in the two intervention groups. In the 106 patients who underwent surgery alone, 22 (21%) patients with positive margins had a 5-year RFS of 45.5%, while those with negative margins had a 5-year RFS of 93.8% and 91.7% at 10 years ($P = 0.00$). However, in the remaining 44 patients who had surgery combined with adjuvant therapies, 22 (48.9%) patients with positive margins had a 5-year RFS of 64.7%, compared with 71.5% in those with negative margins ($P = 0.88$, Fig. 4).

For different anatomical subgroups, positive margin was also a significant prognostic factor for extra-abdominal tumors, but not abdominal wall tumors. Patients with a positive margin of extra-abdominal tumors, had a 5-year RFS of 47% (95% CI, 27–67%) that was unchanged at 10 years, whereas those with negative margins had a 5-year RFS of 82% (95% CI, 70–93%; $P = 0.002$) that was unchanged at 10 years. The 5-year RFS in abdominal tumor patients with positive margins was 78% compared to 93% in those with negative margins ($P = 0.37$).

The ratio of positive margins by anatomic site was 14.7% (11/75) in the abdominal wall, 47% (8/17) in the chest wall and back, 42.4% (14/33) in extremities or girdles, and 42.3% (11/26) in head and neck. Notably, the different outcome by anatomic site was not correlated with an increased in positive margins.

Adjuvant Radiation Therapy

The selective use of adjuvant radiation was delivered in 25 patients. Patients with radiation therapy had a 5-year RFS of 63% (95% CI, 42–84%) that was unchanged at 10 years, whereas patients without adjuvant radiation had a 5-year RFS of 82% (95% CI, 75–89%) and a 10-year RFS of 81% (95% CI, 74–88%; $P = 0.06$).

In total, 7 (7%) of 107 patients with negative margins and 18 (41%) of 44 with positive margins received adjuvant radiation. Patients with positive margins who received adjuvant radiation had a slightly better outcome than those who did not, at 60% versus 51% respectively, although this was not statistically significant ($P = 0.26$) (Fig. 5).

DISCUSSION

DTs are rare and remain difficult to treat due to their infiltrative growth, high propensity for LR and variable natural history. Although several prognostic factors for LR have been identified in the literature, discrepancies still exist and treatment recommendations for these tumors remain contradictory. In our study, we examined a series of 151 patients with extra-abdominal and abdominal wall DTs,

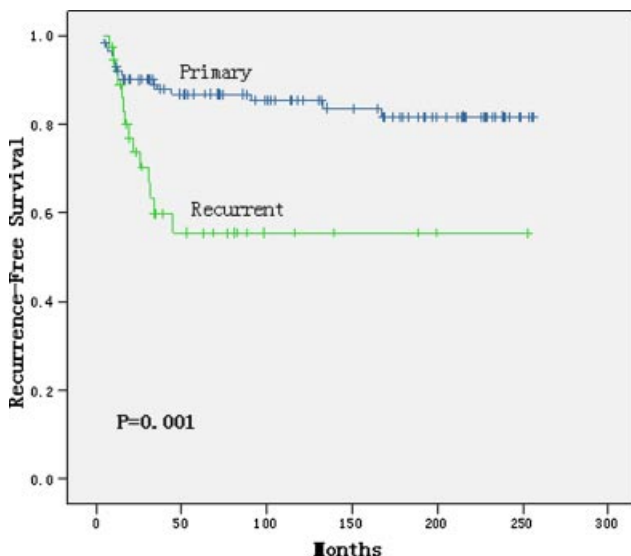


Fig. 1. Local recurrence free survival (RFS) by admitted status in our institution (113 primary cases and 38 recurrences). Tick marks overlapping survival curves denote censored times. [Color figure can be viewed in the online issue, available at www.interscience.wiley.com.]

TABLE III. Factors Potentially Affecting Local RFS for Primary and Recurrent Disease

Factors	Primary			Recurrent		
	Number of patients	10-year RFS (%)	P-value	Number of patients	10-year RFS (%)	P-value
Gender			0.190			0.469
Male	19	79		13	41	
Female	94	87		25	64	
Size			0.013			0.051
≤5 cm	61	95		12	82	
>5 cm	52	73		26	41	
Location			0.000			0.568
Extra-A	50	72		26	58	
Abdominal	63	96		12	52	
Number			0.263			0.015
Single	110	86		27	68	
Multicentric	3	0		11	26	
Margin			0.00			0.09
R0	85	92		21	71	
R1	28	64		17	35	

Extra-A, extra-abdominal.

who had undergone macroscopically complete resection in a single institution over a 20-year time period. We were able to identify admission status, gender, tumor sites, number, size and margin status to be factors predictive of LR in univariate analysis. Size and margin status were also independent prognostic factors in the multivariate analysis.

Some authors have reported an increased risk of LR in female patients and in patients older than 30 years [8], while others have shown risk of LR in patients younger than 30 years [9,10]. However, most other studies were unable to show a correlation between LR and the gender and age of the patients. In our series, female patients had a significantly better outcome than did male patients. However this discrepancy was not apparent in subgroup analysis or in a multivariate setting.

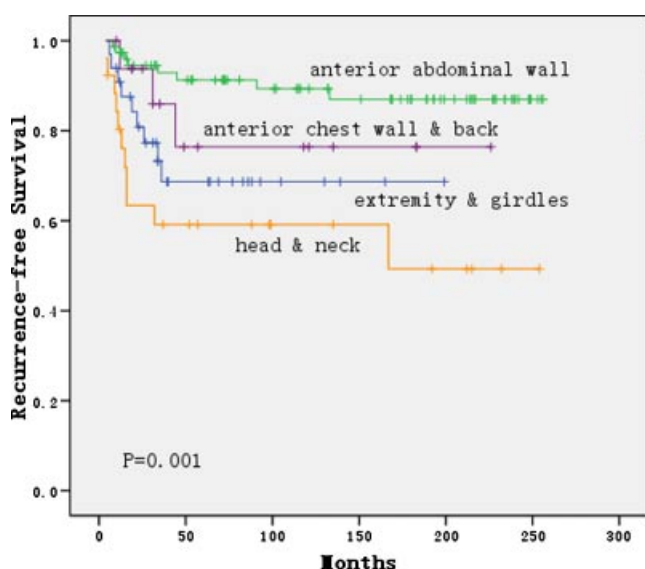


Fig. 2. Local recurrence (LR) free survival for the entire group by location are shown (n=151). Tumors situated in extremities/girdles and head/neck significantly increase the risk of LR compared with those in abdominal wall ($P=0.02$, $P=0.00$ respectively). [Color figure can be viewed in the online issue, available at www.interscience.wiley.com.]

Tumor size with a cut-off point of 5 cm was found to be an independent prognostic factor for primary disease, which was also confirmed by Gronchi et al. [11] and Lev et al. [12]. In contrast, Posner et al. [9] and Merchant et al. [13] did not show this correlation in their series. Patients presenting with multifocal disease tended to have aggressive disease and high risk of LR [14]. In our series, 14 (9%) of 151 patients with multifocal tumors obtained an extremely low 5-year RFS of 18.4% and had an increased risk of LR, as determined by univariate analysis.

Regarding tumor classification, we specifically separated the abdominal wall type from extra-abdominal DTs as a unique type for evaluation. A difference in location of tumor between men and women was found in our study. In the extra-abdominal group, the male to female ratio was approximately 1:1.8, whereas a predominance of female in the fertile and middle ages was noted in the abdominal wall group, at 70 (93%) women out of 75 patients. This has also been confirmed in many other articles [15,16]. With respect to RFS, the abdominal wall group had better outcomes than the extra-abdominal group in univariate analysis ($P=0.001$), however, this difference was not apparent in multivariate analysis. For the further subgroup analysis of extra-abdominal patients, the worst prognosis was seen in the head and neck group, followed by extremities/girdles group, while the outcome of chest wall and back group was equal to that of the abdominal wall group. However, the ratio of positive margins was similar among the extra-abdominal group, which implied that this outcome discrepancy was not correlated with an increase in positive margins. Of note, patients with head and neck tumors suffered the lowest 5-year RFS of 27.3% with positive margin, which was not apparent in the negative margin group ($P=0.001$). We therefore consider that this may mainly be attributed to the contiguousness of neurovascular structures, which compromise the achievement of negative margins and thus the eradication of tumors in the neck area or extremities.

The optimal treatment protocol for DTs is still in dispute. Because of its infiltrative growth and high propensity for LR, surgery remains the treatment mainstay for this tumor, if feasible. However, the value of a positive margin remains controversial. Actually, the precise significance of margin status on LR is difficult to evaluate, as most reports included either a small number of patients, mixing intra-abdominal type as well as recurrent and primary disease, or included patients who had received multiple forms of adjuvant treatment.

Nuytens et al. in 1999, reviewed 22 articles on DTs, finding that margins are of a statistically significant importance [17]. Spear et al.

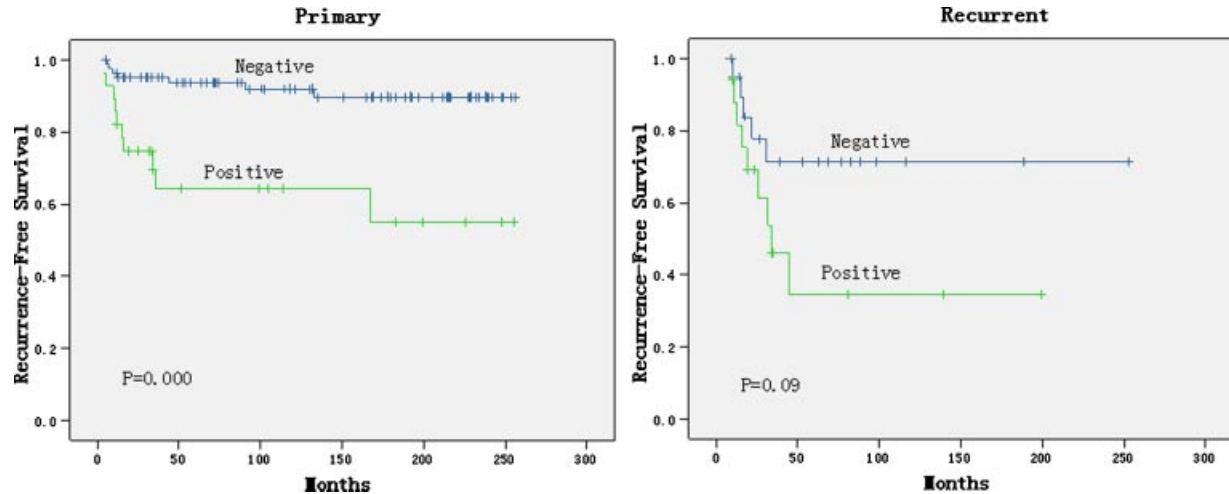


Fig. 3. Local RFS of primary cases (113 patients) and LR free survival (RFS) of recurrent diseases (38 patients) by marginal status. Tick marks overlapping survival curves denote censored times. [Color figure can be viewed in the online issue, available at www.interscience.wiley.com.]

[18] demonstrated that microscopically positive margins significantly influenced LR in a multivariate analysis (19% negative vs. 39% positive). Similar positive results were obtained by small studies such as Posner et al. [9] in their multivariate analysis of 138 patients and Goy et al. [14] in a series of 68 patients. However, their studies might be less numerous and much less selected. Merchant et al. [13], and Gronchi et al. [11], and Lev et al. [12] as recently as 2007 failed to demonstrate this type of impact. Although these reports were better selected and formed large series, they were again either mixed anatomic tumor sites or included patients with surgery and combined adjuvant therapies.

A relatively critical comparative analysis by Leithner et al. [19] concluded that wide or radical excision was the treatment of choice. In our series, positive margin was an independent prognostic factor that significantly influenced LR in the multivariate analysis (11% negative vs. 42% positive). Moreover, patients were clearly classified by anatomic site, and admission status, as well as by therapeutic strategies for further analysis. The significant difference was found mainly in patients with primary disease (8% negative vs. 36%

positive), and extra-abdominal location (16% negative vs. 49% positive), as well as patients who underwent surgery alone (9% negative vs. 52% positive). It is concluded that adjuvant therapies might have alleviated the adverse impacts of positive margins, which were evident in patients undergoing surgery alone. For recurrent disease, margin status showed a trend for predicting local progression, although not statistically significant. However, since all of these studies including ours were retrospectively analyzed, it is unlikely that biases would have been avoided, regardless of how strictly the groupings were done or analyzed. What we can conclude from our retrospective results is that, if feasible, microscopically negative margins should always be the goal of surgery, especially for extra-abdominal tumors. Adjuvant therapies might alleviate the adverse impact of positive margins on LR.

The role of radiotherapy in the management of DT is also controversial. Some authors have demonstrated radiation therapy to be effective in local control, both in adjuvant and primary settings [8,20,21], as well as demonstrating that the adverse effect of positive resection margins would be offset by the addition of radiation [17,18,22]. In contrast, others have shown little benefit in the use of

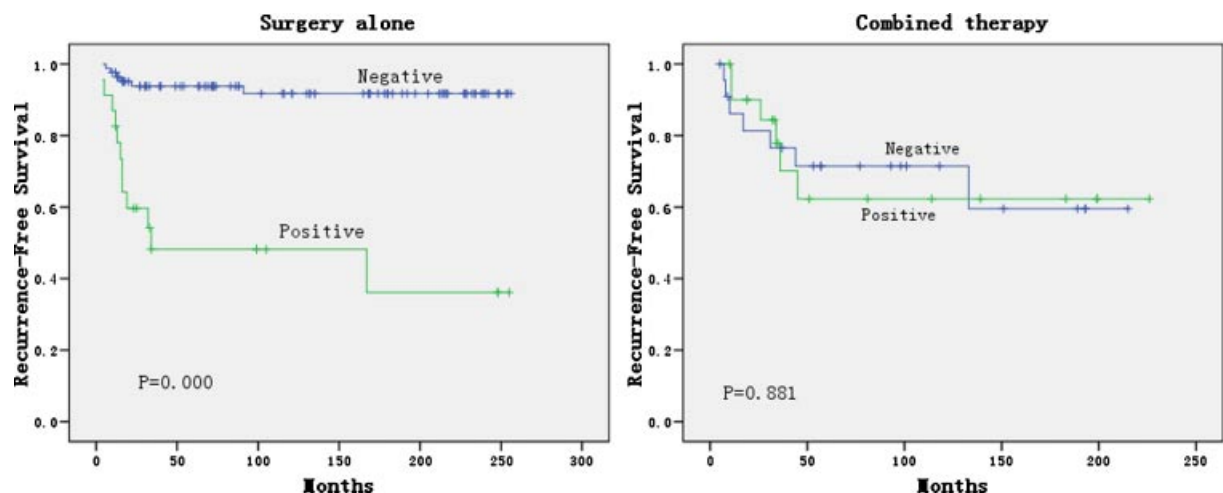


Fig. 4. Margin status significantly affected LR in surgery alone group (106 patients). However, the impact was not significant in combined therapies group. [Color figure can be viewed in the online issue, available at www.interscience.wiley.com.]

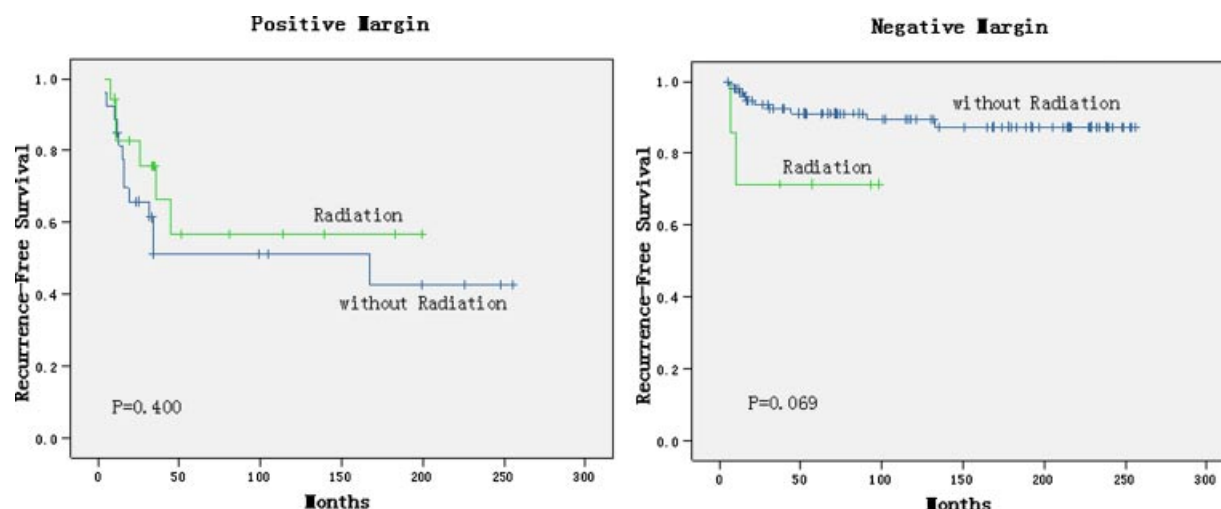


Fig. 5. Effects of radiotherapy and margin status are shown. Twenty-five (17%) selectively received adjuvant radiation. Eighteen patients (40%) with positive margin and radiation had a better RFS than those without radiation, though not statistically significant. Seven patients (7%) with negative margin received radiation and had a worse RFS compared with those without radiation. [Color figure can be viewed in the online issue, available at www.interscience.wiley.com.]

radiation [23] and sometimes worse results with radiation therapy than with surgery alone [8,23,24]. Our study did not show a decrease in the LR rate following selective application of adjuvant radiotherapy (32% with vs. 18% without). In addition, one patient developed a radiation induced sarcoma (MPNST) about 10 years after radiation. Although patients with positive margins and adjuvant radiotherapy had a slightly decreased LR than those without radiotherapy, the addition of radiation therapy on a positive margin did not statistically improve survival. However, all of the series were conducted on a relatively small number of patients in a retrospective setting, with a variation in radiation energy resource and doses across the studies. Criteria for adjuvant radiotherapy use were also not specified [12]. It is difficult to evaluate the role of radiotherapy without a prospective evaluation of standardized adjuvant radiotherapy with large number of patients.

The effect and application of systemic therapies is also controversial. Pharmacological agents are usually the initial agents of choice when managing desmoids in FAP. Hansmann et al. [25] reported their successful experience with high doses of tamoxifen and sulindac as a first-line treatment for FAP-associated tumors, achieving a 90% local control rate, which was <40% in sporadic DTs. In our series, we administered tamoxifen at 30 mg/day in 25 selected patients who had a local control rate of 67% after complete resection. However, no further improvement in local control was revealed. Imatinib mesylate has been reported to exert effective local control in the salvage management of unresectable DTs by several authors [26,27] and should be added as an additional tool for systemic therapies, and this requires further evaluation.

Considering the variable natural history of DTs, with some appearing to be stable or spontaneously regressing, even in the absence of treatment, the emerging application of a wait-and-see policy is also an area of controversy. Lewis et al. [28] from MSKCC reported six patients confronting amputation who had undergone observation only. None experienced disease progression and three experienced some spontaneous regression. Bonvalot et al. [29] from IGR in France demonstrated that patients with primary disease who had undergone microscopically complete surgery had similar outcomes to patients in a no-surgery group (3-year EFS of 65% vs. 68%). In our series, five patients developed recurrent disease after macroscopically resection and choose observation. One of these experienced progression, and the remaining obtained stable disease or tumor regression.

In conclusion, at this point in time, functionally preserving surgery remains the mainstay for treatment of DTs. Concerning that spontaneous growth arrest is not an uncommon feature of this disease and that metastasis rarely occurs, a wait-and-see policy may also be applied in some circumstances of primary disease or for unresectable disease. However, this does not mean that we can compromise or neglect a positive margin. Our experience, as in other studies, is limited, as it is based on retrospective analysis. Prospective randomized trials are urged for resolving the questions that are currently in dispute to allow the design of a rationale therapy protocol for desmoid patients.

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